

YASH AMBARRAJ BAGAL

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PROFESSIONAL SUMMARY

Backend Developer with hands-on experience in Java Spring Boot, RESTful API development, and microservices architecture. Skilled in JWT authentication, Spring Data JPA, Hibernate ORM, and relational database design using MySQL and PostgreSQL. Builds AI-integrated systems including LLM-powered code review bots and local voice assistants using Rust, Ollama, and Whisper.cpp. Experienced in building scalable, maintainable enterprise applications following clean architecture and Agile development practices. Active open source contributor with successfully merged pull requests.

TECHNICAL SKILLS

Programming Languages: Java, Rust, Python, JavaScript, TypeScript, SQL, C++

Backend Frameworks & Libraries: Spring Boot, Spring MVC, Spring Data JPA, Hibernate ORM, Node.js

Databases: MySQL, PostgreSQL

AI & LLM Tools: Ollama, Whisper.cpp, Piper-TTS, LLM Inference, NLP, GitHub API Integration

Tools & DevOps: Docker, Git, GitHub, Maven, Postman, GitHub Actions

Core Competencies: RESTful API Design, Microservices Architecture, JWT Authentication, ORM Design Patterns, Database Schema Design, System Design, Agile Methodology

WORK EXPERIENCE

Slytherin EduTech Pvt. Ltd. | Full Stack Developer Trainee (Remote) *Aug 2024 – Jan 2025*

- Developed and deployed modular RESTful APIs using Java Spring Boot for multiple enterprise web applications, improving system modularity and reducing development time.
- Implemented JWT-based authentication and role-based access control (RBAC) to secure API endpoints and manage user authorization across multiple roles.
- Built robust backend services using Spring Data JPA and Hibernate ORM for efficient database operations, data persistence, and complex entity relationship management.
- Collaborated within an Agile development team, participating in sprint planning, code reviews, and following best practices for code quality and maintainability.

Arcelor Private Limited | Software Development Intern (Remote) *Jun 2023 – Jul 2023*

- Developed backend APIs for internal business dashboards using PHP, handling core business logic and data processing workflows.
- Implemented server-side functionality for MySQL database operations, dynamic content management, and data retrieval for reporting features.

TECHNICAL PROJECTS

PR Review Bot | *Python, GitHub API, LLM, Ollama, Async Processing* *2025*

- Built an intelligent automated code review system that integrates with GitHub webhooks to trigger real-time pull request analysis on PR open, update, or reopen events.
- Implemented a dual analysis engine combining fast static heuristics (secret detection, null checks) with deep contextual LLM inference using a locally hosted Ollama model for privacy-preserving code review.
- Engineered GitHub-native integration that publishes structured review comments directly on pull requests, enabling seamless developer feedback within existing workflows.
- Designed an extensible rule-based architecture for adding new heuristic checks, with production-ready async processing, comprehensive error handling, and structured logging.

RustAssistant | *Rust, Whisper.cpp, Ollama, Gemma LLM, Piper-TTS, Async Rust* *2025*

- Developed a fully local voice assistant in Rust that performs end-to-end voice-to-response-to-speech processing with zero cloud dependencies, ensuring complete user privacy.

- Integrated Whisper.cpp for accurate real-time speech-to-text transcription and Piper-TTS for high-quality natural speech synthesis as the audio output layer.
- Leveraged Ollama with Gemma 3.x LLM models for intelligent conversational AI responses, running entirely on local hardware with optimized async Rust runtime for low latency.
- Architected a modular async pipeline using Rust's async runtime to coordinate speech recognition, LLM inference, and TTS synthesis with minimal response latency.

Daily Diary AI | Java, Spring Boot, PostgreSQL, NLP, AI/ML

2025

Repository: <https://github.com/yash1648/Daily-Diary-AI>

- Engineered a backend system integrating Natural Language Processing (NLP) for AI-powered emotional analysis and mood detection from daily journal entries.
- Designed RESTful APIs for journal entry management with encrypted data storage, ensuring privacy-first data handling and cloud synchronization.
- Developed backend services for mood pattern detection, historical trend analysis, and AI-driven personalized insights delivery to users.
- Implemented secure user authentication and encrypted cloud-synced workflows to ensure confidentiality and data integrity.

Library Management System | Java, Spring Boot, Hibernate ORM, MySQL, JWT

2025

Repository: <https://github.com/yash1648/Library-Management-System>

- Designed and developed a comprehensive RESTful API backend using Spring Boot and Hibernate ORM to manage books, users, and library transactions.
- Implemented JWT-based authentication system with role-based access control (RBAC) supporting multiple user types including admin, librarian, and member roles.
- Created normalized relational database schema with properly defined entity relationships and constraints for books, users, and borrow/return transactions.
- Built complete CRUD operations for book inventory management, user registration, and borrowing/return workflow automation.

OPEN SOURCE CONTRIBUTIONS

Password Manager | JavaScript, Python, PostgreSQL

2025

Repository: <https://github.com/jaykumarpatil314-ux/Password-Manager>

- Contributed to backend API development with successfully merged pull requests implementing encryption workflows for secure password handling.
- Enhanced existing database operations, optimized query performance, and resolved critical security vulnerabilities following industry best practices.
- Improved backend code architecture and structure, increasing maintainability and security of the password management system.

EDUCATION

R.C. Patel Institute of Technology, Shirpur

2025 – Present

Bachelor of Technology (B.Tech) in Computer Engineering

Nikam Institute of Technology, Dhule

2021 – 2024

Diploma in Computer Engineering

Aggregate: 89% | First Class with Distinction | Ranked 1st in Class

CERTIFICATIONS & ACHIEVEMENTS

- Java Programming Certification – HackerRank (Certificate ID: b527c2e815dd)
- C++ Programming Certification – Udemy (Certificate ID: UC-89e00bb8)
- Rust Programming Certification – CodeChef (Certificate ID: ba36950)
- Winner – "C Mania" Programming Competition
- Open Source Contributor with successfully merged pull requests on GitHub